Azure Sentinel Honeypot Home Lab

Project Overview

This project demonstrates how a cloud-based honeypot was deployed using **Microsoft Azure** and monitored using **Microsoft Sentinel** to analyze real-world RDP brute-force attacks.

Objectives

I. Deploy a vulnerable VM in Azure
II. Collect security logs using Log Analytics
III. Detect failed login attempts (Event ID 4625)
IV. Visualize global attack activity in Sentinel

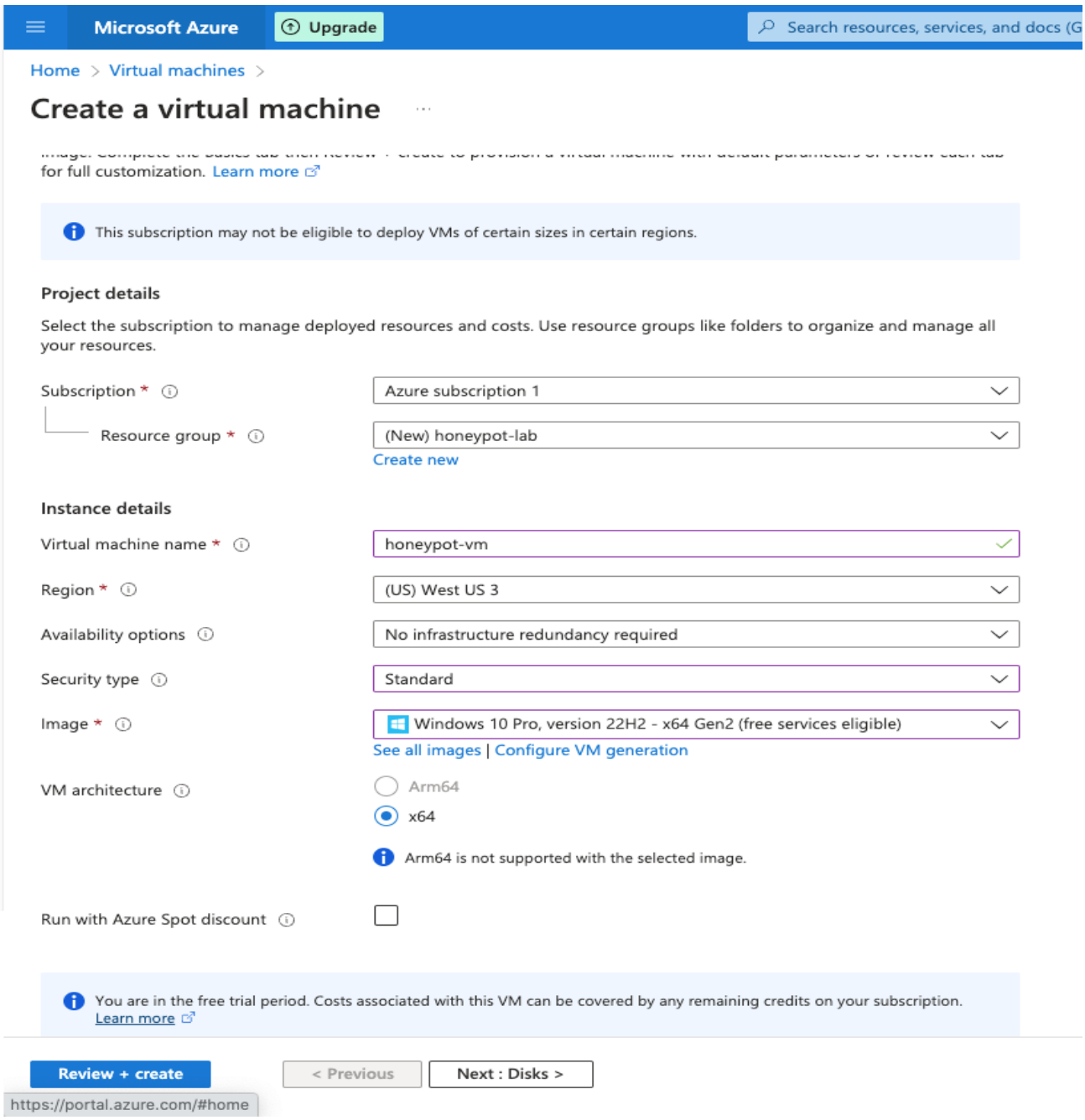
Technologies Used

I. Microsoft Azure
II. Microsoft Sentinel
III. Log Analytics Workspace
IV. PowerShell
V. Remote Desktop Protocol (RDP)
VI. IPGeolocation API
VII. KQL

Lab Architecture

VM → Log Analytics → Sentinel → Attack Map

1. **Setup our honey pot virtual machine**



2. Create a Log Analytics Workspace

Home > Log Analytics workspaces >

Create Log Analytics workspace

Basics | Tags | Review + Create

A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. Learn more

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure subscription 1

Resource group * : honeypot-lab

Create new

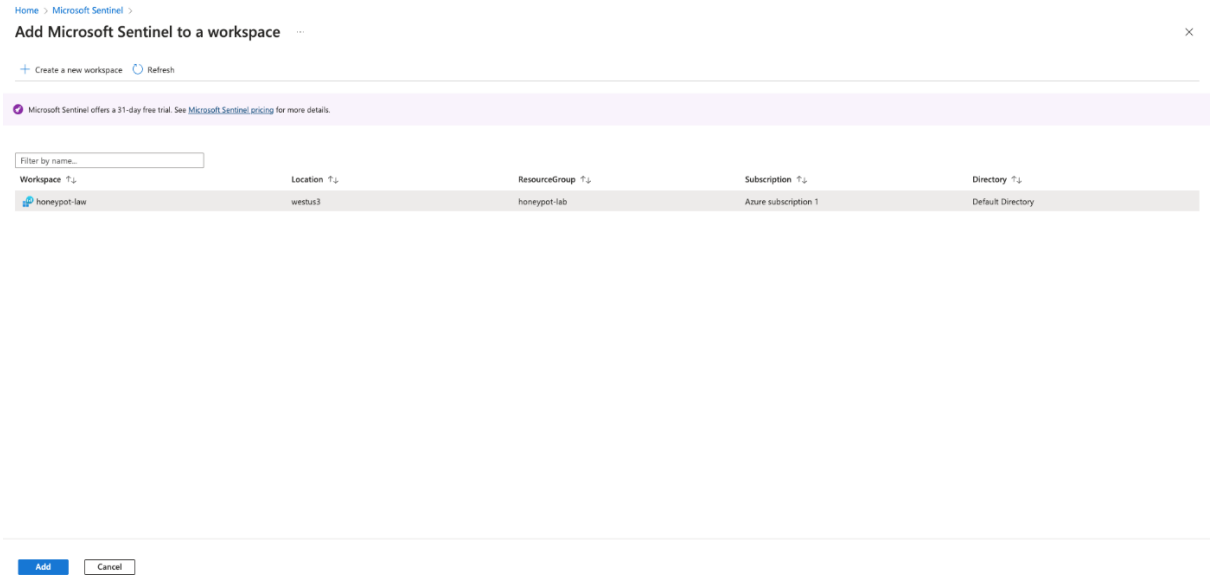
Instance details

Name * : honeypot-law

Region * : West US 3

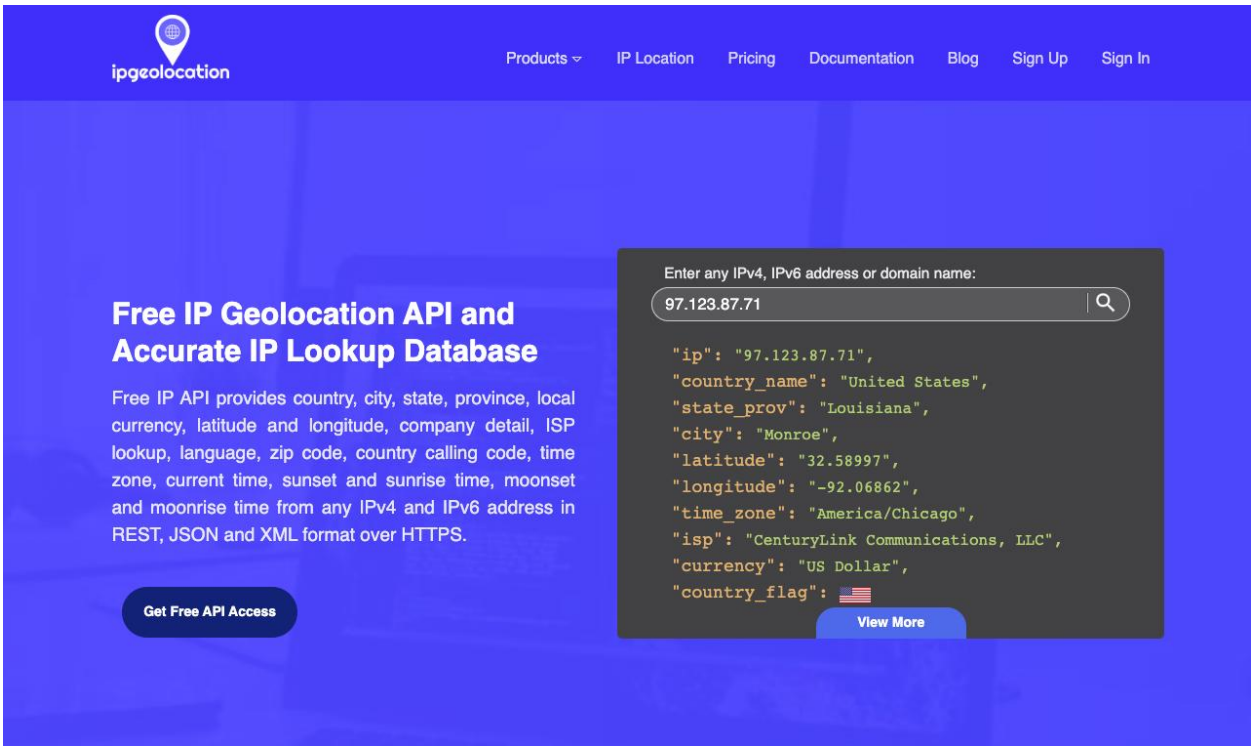
Review + Create | « Previous | Next : Tags >

3. Set up Microsoft Sentinel



4. Attack Map

Data will be exported from Windows Event Viewer and imported into the IP Geolocation service by the script.



Implementation Steps:

a. **Created Azure VM honeypot**

A vulnerable Windows virtual machine was deployed in Microsoft Azure to act as a honeypot system exposed to the internet.

Configuration details:

VM Name: Honeypot-VM
Region: West US 3
Operating System: Windows 10 Pro (22H2)
VM Size: Standard_D2s_v3 (2 vCPUs, 8 GB RAM)
Authentication: Username and password
Inbound Port: RDP (Port 3389) enabled
Resource Group: Honeypot-Lab

The VM was intentionally configured to allow remote connections in order to attract automated brute-force login attempts from external threat actors**.**

Disks

Select Next : Networking >

Networking (Configured Network Security Group)

NIC network security group: Advanced -> Create new

A Network Security Group (NSG) in Azure is a virtual firewall that filters and controls network traffic to protect Azure resources.

By clicking the three dots, delete Inbound rules (1000: default-allow-rdp)
Add an inbound rule
Destination port ranges: * (wildcard for anything)
Protocol: Any
Action: Allow
Priority: 100 (low)
Name: Anything (allow-any-inbound)
Select Review + Create

b. Enabled Cloud Security Monitoring

Security monitoring was configured using Microsoft Defender for Cloud to collect system activity and security events.

c. Connected VM to Log Analytics Workspace

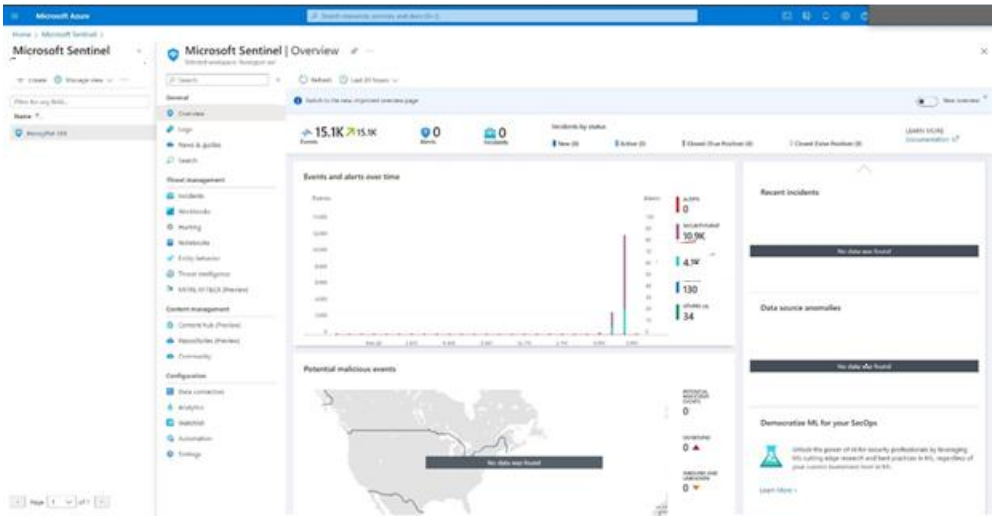
The honeypot VM was connected to an Azure Log Analytics Workspace to centralize log collection and enable security event analysis

d. Enabled Sentinel SIEM

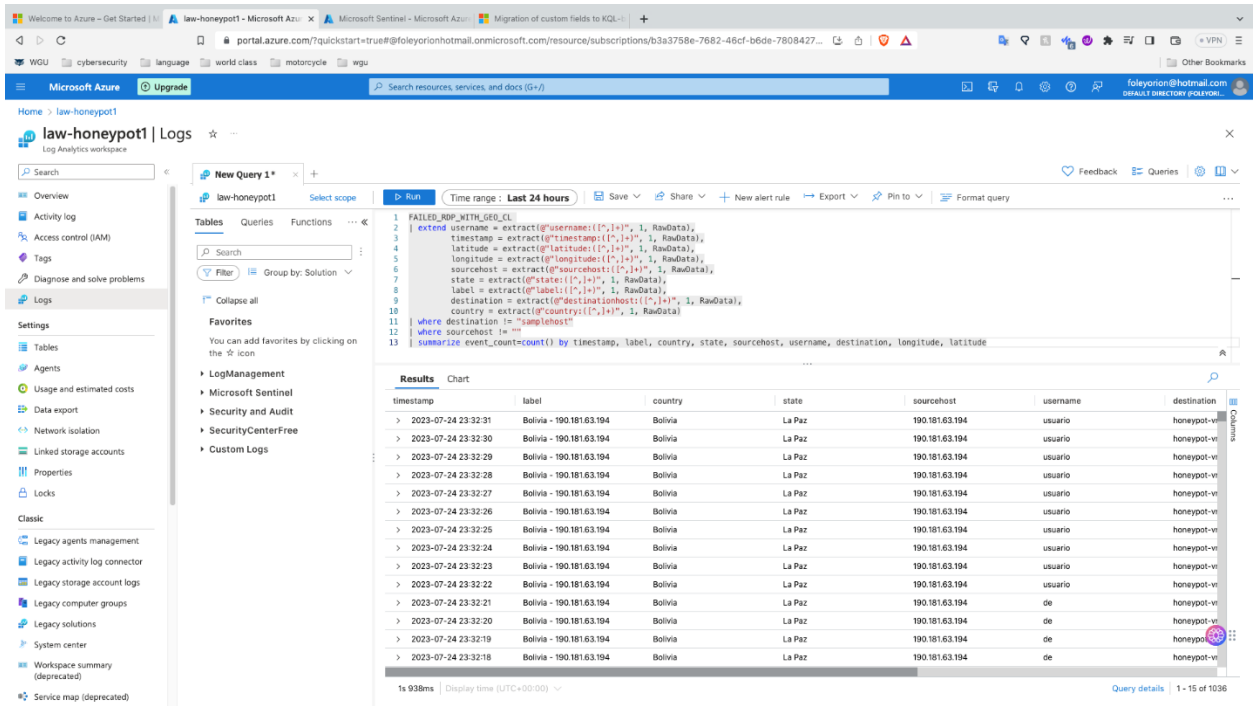
The workspace was integrated with Microsoft Sentinel to analyze security events, detect failed RDP login attempts, and visualize attack activity using KQL queries and Sentinel Workbooks

Attack Visualization

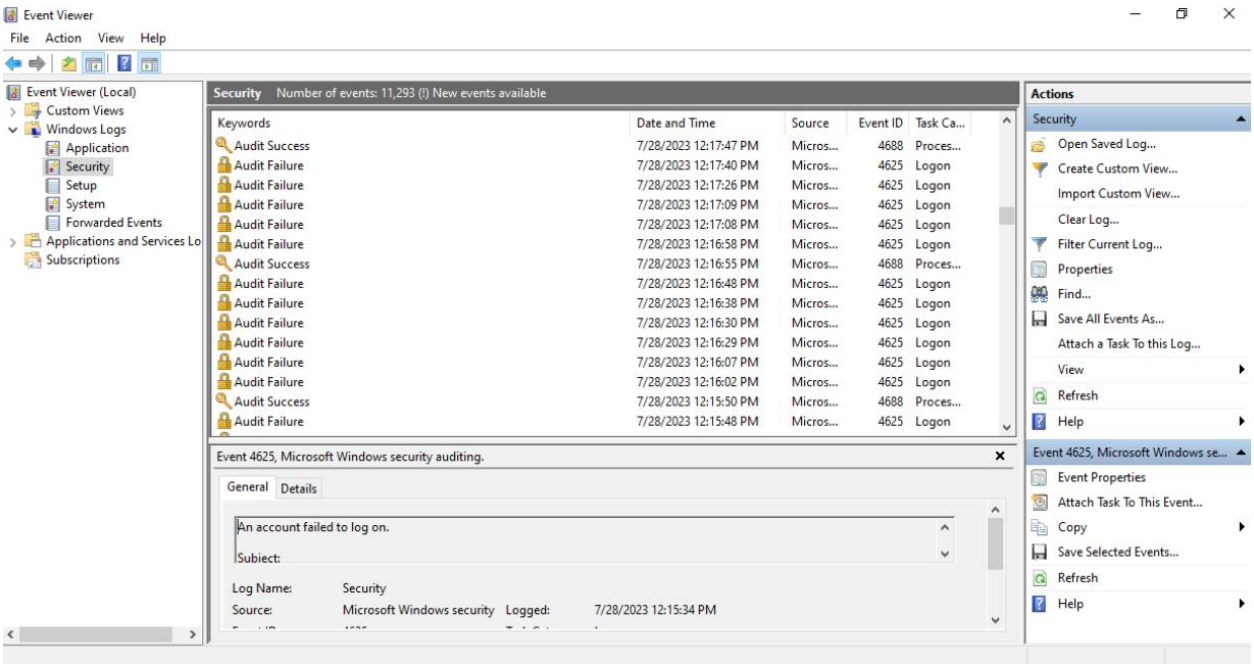
Sentinel dashboard



KQL query results



Event Viewer showcasing failed RDP logon efforts. Event ID: 4625



Attack world map

Layout Settings

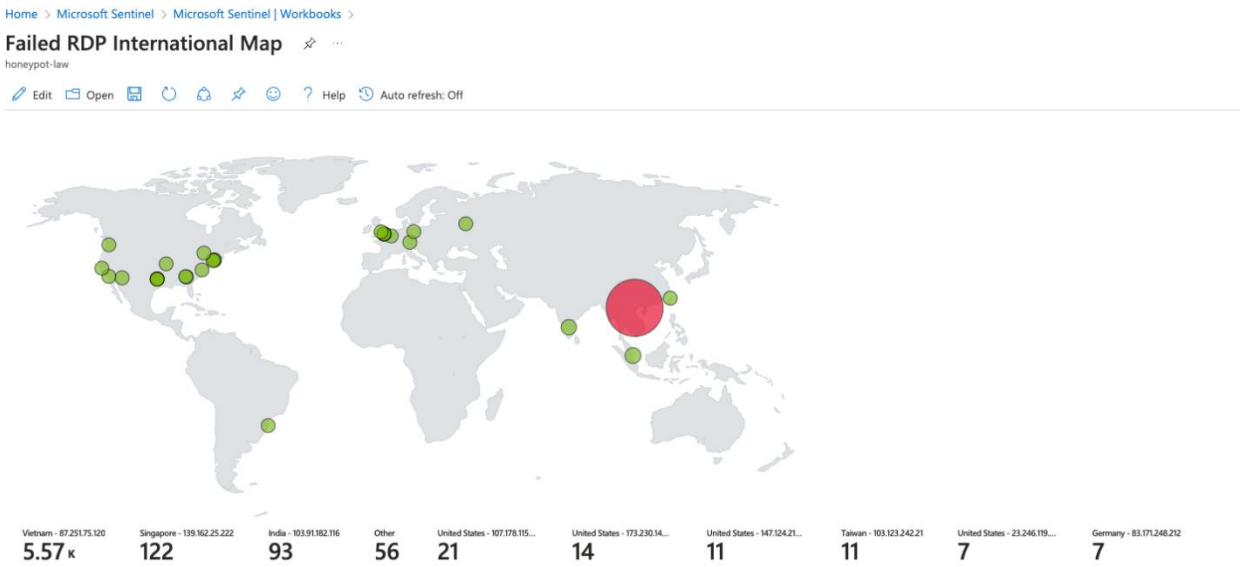
Location info using: Latitude/Longitude
Latitude: latitude
Longitude: longitude
Size by: event_count

Color Settings

Coloring Type: Heatmap
Color by: event_count
Aggregation for color: Sum of Values
Color palette: Green to Red

Metric Settings

Metric Label: label
Metric Value: event_count
Click Apply button and Save and Close
Save as "Failed RDP International Map" in the same region and under the resource group (honeypot-lab)
Keep refreshing the map to show more inbound failed RDP attacks



Results

Thousands of brute-force attempts detected
Attack sources from multiple countries
Automated password guessing activity observed

Conclusion

This project provided practical experience in implementing cloud security monitoring and understanding real-world cyber threats. By deploying a honeypot virtual machine in Microsoft Azure, I observed how systems exposed to the internet quickly attract automated attack attempts.

Using **Microsoft Sentinel,** I learned how Security Information and Event Management (SIEM) platforms collect, correlate, and analyze security logs from multiple sources. The SIEM environment helped visualize security events, detect suspicious login attempts, and monitor attacker activity targeting the honeypot VM.

Log data collected through **Azure Log Analytics** demonstrated the importance of centralized log management. By analyzing event logs and failed authentication attempts, it became possible to identify patterns of brute-force attacks and understand how attackers probe exposed systems.

This project also highlighted the importance of cloud security tools such as **Microsoft Defender for Cloud**, which help detect vulnerabilities, monitor threats, and improve the overall security posture of cloud environments.

Overall, the project strengthened my understanding of **SIEM** monitoring, threat detection, and log analysis, while demonstrating how cloud security services can be used to detect and respond to potential cyber threats in real time.